CSCI 432 Handout 12: Graph Algorithm Recap

Name: ________________________________

March 31, 2026

Shortest Paths

Input: A (directed or undirected) weighted graph such that all edge weights are positive, $G = (V, E, \omega)$, where $\omega \colon E \to R_+$, a start node s, and an end node t.

Output: A shortest path between s and t.

1. Give an example of a graph such that the shortest path IS NOT unique..

   Answer

2. Give an example of a graph such that the shortest path IS unique..

   Answer

3. Give an example of a graph for which there is no path between s and t.

   Answer

Dijkstra's Algorithm

Let's remember how Dijkstra's algorithm works.

minimum

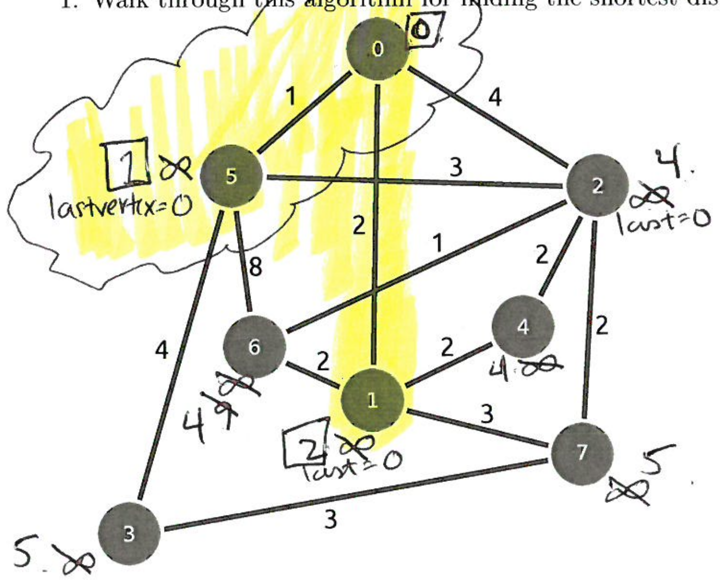
Initial set-up: We start create a priority queue where each vertex has priority ∞, except s, which has priority 0.

The main loop: We pop off the next element x of the priority queue, as we now know it's distance to the start s. We then perform the 'relaxation' step of updating all priorities to allow for going through x. If going through x is shorter than tentative distance ($priority(x) + w(x, v) <$ current priority), then we update the priority.

When t is popped, we return the priority of t.

yes, this is a greedy algorithm!

1. Walk through this algorithm for finding the shortest distance between 0 and 6 in the following graph:



known distances to $s = v_0$.

$d(v_0, v_0) = 0$, path = $[v_0]$

$d(v_0, v_5) = 1$, path = $[v_0, v_5]$

$d(v_0, v_1) = 2$, path = $[v_0, v_1]$

2. What is the runtime of Dijsktra's algorithm? **Answer**

Priority Queue w/ n elts: $\Theta(\log n)$ for insert or pop

Dijkstra's: $\Theta(|V| + |E| \log |V|)$

3. Write psuedo-code so that the algorithm returns a shortest path between s and t, not just the distance. **Answer**